

L6 LiDAR Perception Quiz ANS

1. A LiDAR pulse returns after 1.33 μ s. Using $R = c \cdot \Delta t / 2$, how far away is the target?

- A) About 200 m
- B) About 400 m
- C) About 20 m
- D) About 2 km

ANS: A) About 200 m

2. A return has range $r = 20$ m, azimuth $\theta = 90^\circ$ and elevation $\phi = 0^\circ$. Which Cartesian point (x, y, z) does it map to?

- A) (20, 0, 0)
- B) (0, 20, 0)
- C) (0, 0, 20)
- D) (-20, 0, 0)

ANS: B) (0, 20, 0)

3. Which statement correctly compares pulsed time-of-flight LiDAR with FMCW LiDAR?

- A) Pulsed ToF measures radial velocity directly, while FMCW cannot measure it at all
- B) FMCW measures the round-trip time of a very short laser pulse like a stopwatch
- C) FMCW measures radial velocity directly via the Doppler shift, but is more complex
- D) FMCW is mass-produced today and is cheaper than a pulsed time-of-flight sensor

ANS: C) FMCW measures radial velocity directly via the Doppler shift, but is more complex

4. What is the idea of early fusion of camera and LiDAR, e.g. PointPainting?

- A) Run a camera detector and a LiDAR detector separately and merge their boxes
 - B) Project the points into the image and append the segmentation scores to each point
 - C) Fuse the backbone features of both sensors in one shared BEV feature map
 - D) Replace the LiDAR by a monocular depth estimator that works on the images
- ANS: B) Project the points into the image and append the segmentation scores to each point

5. A LiDAR has an angular resolution of $\Delta\theta = 0.2^\circ$. What is the approximate spacing between neighbouring points at 100 m ($s = R \cdot \tan \Delta\theta$)?

- A) About 0.035 m
- B) About 3.5 m
- C) About 0.35 m
- D) About 0.17 m

ANS: C) About 0.35 m

6. Why do 1550 nm LiDARs allow a higher pulse energy than 905 nm LiDARs?

- A) Light at 1550 nm is not absorbed by water in the atmosphere at all, so power rises
- B) Light at 1550 nm is absorbed before the retina, so the eye-safety limit is higher
- C) Silicon detectors are more sensitive at 1550 nm than they are at 905 nm
- D) The 1550 nm lasers are cheaper and need much less electrical power to run

ANS: B) Light at 1550 nm is absorbed before the retina, so the eye-safety limit is higher

7. Why does the received LiDAR signal in fog or rain fall as $\exp(-2\alpha R)$?

- A) The detector always averages over two consecutive laser pulses before it reports
- B) The symbol α is defined as twice the extinction coefficient of the air
- C) The light travels out and back, so the attenuation acts over a path of $2R$
- D) Rain reflects exactly half of all the pulses back to the sensor immediately

ANS: C) The light travels out and back, so the attenuation acts over a path of $2R$

8. Which failure mode creates 'ghost points' beyond walls or vehicles?

- A) Dark targets that return less than 5 % of the emitted photons
- B) Multipath: the pulse bounces over two surfaces before it returns

C) Retroreflector blooming around road signs and license plates

D) Motion distortion of a spinning sweep at high vehicle speed

ANS: B) Multipath: the pulse bounces over two surfaces before it returns

9. A 128-channel LiDAR has 1,800 azimuth steps and rotates at 10 Hz. At most how many points does it produce per second?

A) About 230,000

B) About 23 million

C) About 2.3 million

D) About 18,000

ANS: C) About 2.3 million

10. A 0.5° rotation error in the LiDAR extrinsic calibration shifts a point at 100 m laterally by about how much?

A) About 0.09 m

B) About 9 m

C) About 0.5 m

D) About 0.9 m

ANS: D) About 0.9 m

11. A spinning LiDAR needs about 100 ms per sweep and the car drives at 20 m/s. What does deskewing do about the resulting 2 m of motion?

A) It discards all the points that were measured in the first half of the sweep

B) It increases the rotation speed of the sensor from 10 Hz to 20 Hz

C) It transforms each point with the vehicle pose at its own timestamp

D) It applies a single pose, taken at the end of the sweep, to all points

ANS: C) It transforms each point with the vehicle pose at its own timestamp

12. An $80\text{ m} \times 80\text{ m} \times 4\text{ m}$ region at 0.1 m voxels has 25.6 M voxels, but a sweep has only $\approx 0.19\text{ M}$ points. What does this motivate?

A) Larger voxels combined with dense 3D convolution over the whole region of interest

B) Using fewer LiDAR channels so that fewer points are produced in every sweep

C) Dropping the height coordinate and using only the raw intensity of each point

D) Sparse convolution (SECOND) or pillars, as dense 3D convolution mostly processes empty space

ANS: D) Sparse convolution (SECOND) or pillars, as dense 3D convolution mostly processes empty space

13. RANSAC fits a ground plane (3 points per sample) with inlier ratio $w = 0.5$ and confidence $p = 0.99$. How many iterations $N = \ln(1 - p)/\ln(1 - w^3)$ are needed?

A) About 35

B) About 17

C) About 169

D) About 4,603

ANS: A) About 35

14. What is the main advantage of DBSCAN over k-means for clustering LiDAR points?

A) It always finds exactly one cluster for each object that is in the scene

B) It is faster than every other clustering algorithm when it is run on a GPU

C) It uses one single ϵ that works equally well at near and at far ranges

D) The number of clusters is not needed, and sparse points are rejected as noise

ANS: D) The number of clusters is not needed, and sparse points are rejected as noise

15. A parked car is rotated by about 30° . Why is a min-area rectangle a better box than an axis-aligned one?

A) The axis-aligned box under-estimates the height of the car that is in the scene

B) The axis-aligned box over-estimates the footprint (15.2 vs. 8.7 m^2) and gives no heading

C) The min-area rectangle can be computed without any prior clustering step at all

D) An axis-aligned box cannot be computed from a LiDAR point cloud in any way

ANS: B) The axis-aligned box over-estimates the footprint (15.2 vs. 8.7 m^2) and gives no heading

16. Why does ICP need a good initial guess, such as odometry?

- A) Nearest-neighbour matching converges only to a local minimum of the alignment error
- B) ICP does not use any point matching and relies only on image features of the scans
- C) ICP can estimate only a translation and can never estimate a rotation of the scan
- D) The SVD solution is valid only for two scans that are perfectly identical

ANS: A) Nearest-neighbour matching converges only to a local minimum of the alignment error

17. Why is PointNet invariant to the order of the input points?

- A) A shared per-point MLP followed by max-pooling, which is a symmetric function
- B) It sorts the points by their range before any of them is processed by the network
- C) It first voxelizes the points into a regular three-dimensional grid of cells
- D) It uses a recurrent network that scans the points of the sweep one by one

ANS: A) A shared per-point MLP followed by max-pooling, which is a symmetric function

18. What is the main design idea of PointPillars?

- A) Dense 3D convolution over a fine voxel grid that covers the whole region around the car
- B) Sparse 3D convolution that touches only the non-empty voxels of a regular grid
- C) Raw point processing with hierarchical sampling and grouping of local neighbours
- D) Pillars drop the z split, so a pseudo-image is processed with fast 2D convolutions only

ANS: D) Pillars drop the z split, so a pseudo-image is processed with fast 2D convolutions only

19. Which evaluation rule is used for KITTI cars, and how does nuScenes differ?

- A) KITTI: $\text{IoU} \geq 0.7$ counts as correct; nuScenes matches by center distance (0.5 to 4 m)
- B) KITTI matches by center distance, while nuScenes uses an IoU threshold of 0.7 for cars
- C) Both count only the number of points that fall inside the predicted bounding box
- D) Both use classification accuracy without any measure of the localization error

ANS: A) KITTI: $\text{IoU} \geq 0.7$ counts as correct; nuScenes matches by center distance (0.5 to 4 m)

20. Why does the lecture stress worst-case rather than average latency, and what does 100 ms mean at a 60 m/s closing speed?

- A) The average latency is always larger than the worst case; 100 ms is 0.6 m
 - B) Only the throughput of the system matters for safety; 100 ms is 60 m
 - C) Latency is irrelevant at a 10 Hz sweep rate; 100 ms corresponds to 1 m
 - D) One late result can be unsafe, so the maximum matters; 100 ms at 60 m/s is 6 m
- ANS: D) One late result can be unsafe, so the maximum matters; 100 ms at 60 m/s is 6 m